

Laboratory of Applied & Industrial Electronics

#3 Mixer

Lecturer: Prof. Daniele Scirè



**Università
degli Studi
di Palermo**



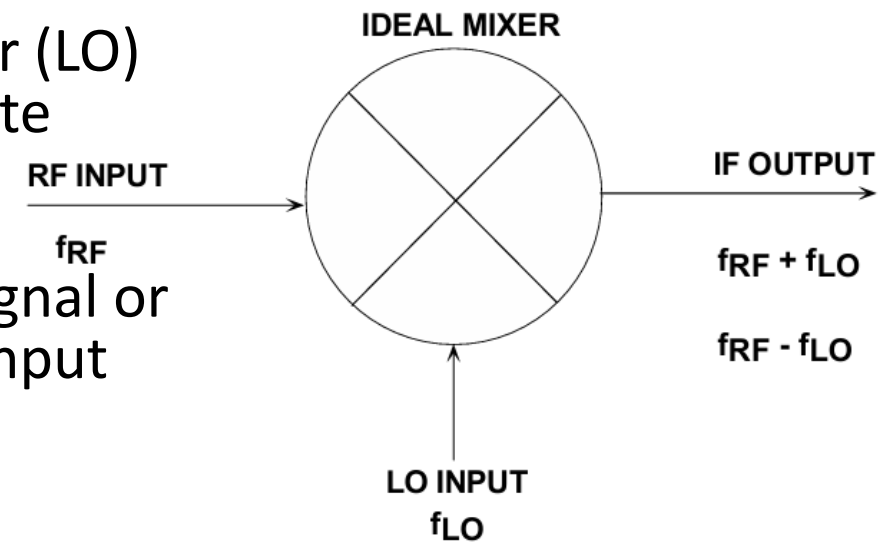
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Introduction

- One of the most useful radio frequency processes is that of mixing.
- It differs from the audio mixer where signals are simply added together. Here signals are multiplied together and a new signal at a different frequency is generated.
- Such a process is non-linear otherwise we could not obtain a different frequency.
- The mixing or multiplication is used in virtually every communication device since it is integrated in both transmitter and receiver.

Basic concepts

- A mixer is a 3-port electronic circuit. It has two input ports and an output port. The ideal mixer process the input signals such that the output signal frequency is the sum (or difference) frequency of the inputs.
- The nomenclature for the 3 ports are the Local Oscillator (LO) port, the Radio Frequency (RF) port, and the Intermediate Frequency (IF) output port.
- The LO port is typically driven with either a sinusoidal signal or a square wave signal. The LO port is usually used as an input port.
- The other 2 ports of the mixer, the RF and IF, can be interchanged as either the second input or the output. The actual configuration depends on the application.

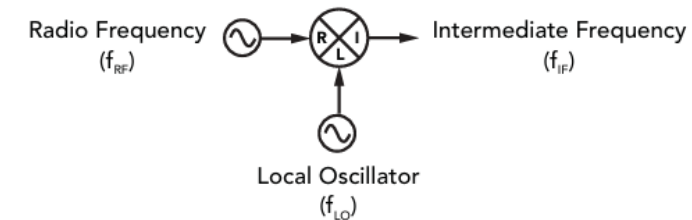
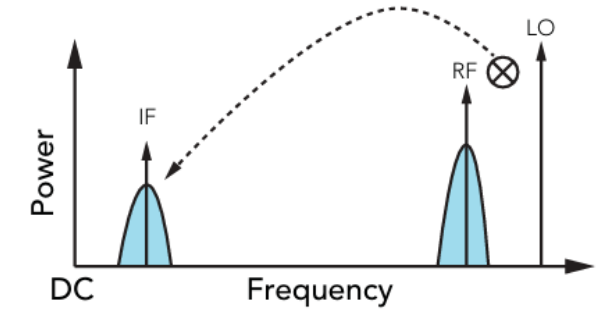


Basic concepts

- When the desired output frequency is lower than the second input frequency, then the process is called downconversion and the RF is the input and the IF is the output (the sum frequency is called image response).
- Differently, when the desired output frequency is higher than the second input frequency, then the process is called upconversion and the IF is the input and the RF is output.
- For the upconversion case we have the proximity of the sum and difference frequencies (f_{RF1} and f_{RF2}) and both are available at the RF output port. The single sideband can be selected through a bandpass filter.
- IF/RF signals carry out the information. During the frequency conversion, the information carried by the signal is frequency translated to the output. Therefore, mixers perform the critical function of translating in the frequency domain.

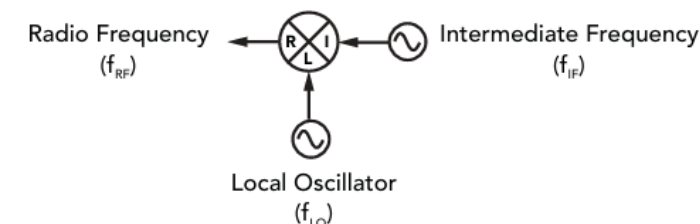
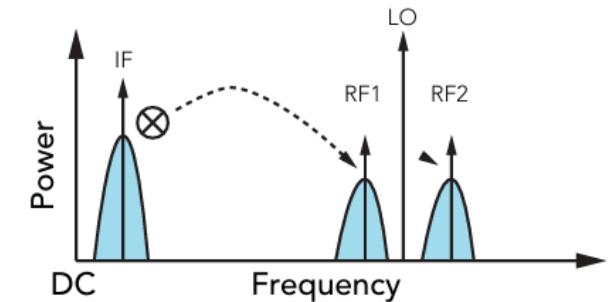
DOWNCONVERSION

$$f_{IF} = |f_{LO} - f_{RF}|$$



UPCONVERSION

$$f_{RF1} = f_{LO} - f_{IF} \quad f_{RF2} = f_{LO} + f_{IF}$$

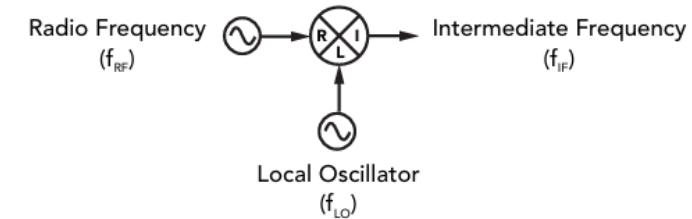
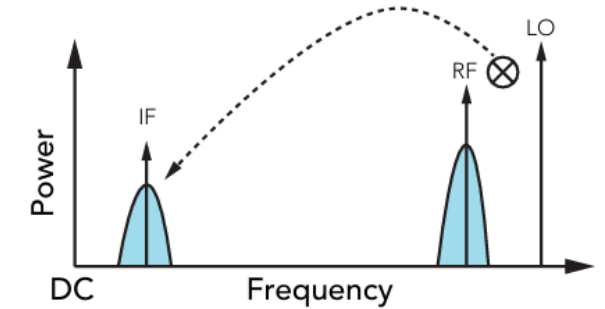


Basic concepts

- Each of the outputs (both in downconversion and upconversion) is only half the amplitude (one-quarter the power) of the individual inputs.
- In dB this means that there is a loss of 6 dB. In a practical multiplier, the conversion loss may be greater.

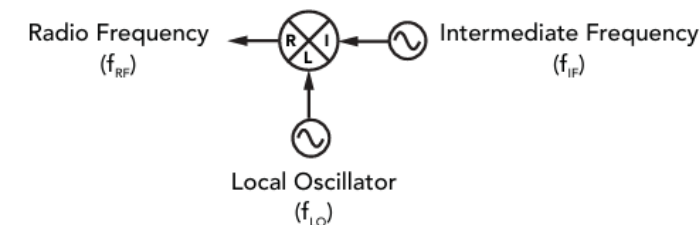
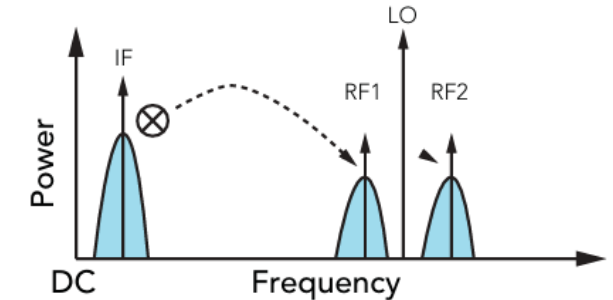
DOWNCONVERSION

$$f_{IF} = |f_{LO} - f_{RF}|$$



UPCONVERSION

$$f_{RF1} = f_{LO} - f_{IF} \quad f_{RF2} = f_{LO} + f_{IF}$$



Basic concepts

A mixer can be implemented in several ways, using active or passive techniques.

We can identify three subclasses of circuits, sharing certain similarities. All are in the class of signal multipliers, producing at their output a signal which is, in one way or another, the product of its two inputs: multipliers, modulators, and mixers.

- An **analog multiplier** has two signal input ports, which can be called X and Y , and generates an output W that is the linear product of the voltages applied to these two ports. To retain dimensional consistency, the analog linear multiplication function must invoke the use of a reference voltage, which we can call U , thus $W = XY/U$. In some cases, U is a third input that can be used to implement analog division. In single-quadrant multipliers, X and Y must be unipolar; in two-quadrant multipliers, one of the inputs may be bipolar; in four-quadrant multipliers, both X and Y may be bipolar

Basic concepts

- **Modulators** can be viewed as sign-changers. The two inputs, X and Y , generate an output W , which is simply one of these inputs multiplied by just the sign of the other that is $W = Y \operatorname{sign}(X)$. Therefore, no reference voltage is required.

A good modulator exhibits very high linearity in its signal path, with precisely equal gain for positive and negative values of Y , and precisely equal gain for positive and negative values of X .

- A **mixer** is a modulator optimized for frequency-translation. Its place in the signal path is usually close to the antenna, where both the wanted and (often large) unwanted signals coexist at its signal input, usually called the RF port. Thus, the mixer must exhibit excellent linearity in the sense that its output (at the IF port) is expected to increase by the same number of dB as a test signal applied to the RF port, up to as high as level as possible. This attribute is defined both by the 1 dB gain-compression and the 3rd-order intercept. The conversion process is driven by an input applied to the LO port.

Performance metrics

Noise and matching characteristics are crucial to achieving acceptable levels of performance in a receiver's mixer. It is desirable to keep the LO power to a minimum to minimize cross-talk between the three ports, but this often conflicts with other requirements.

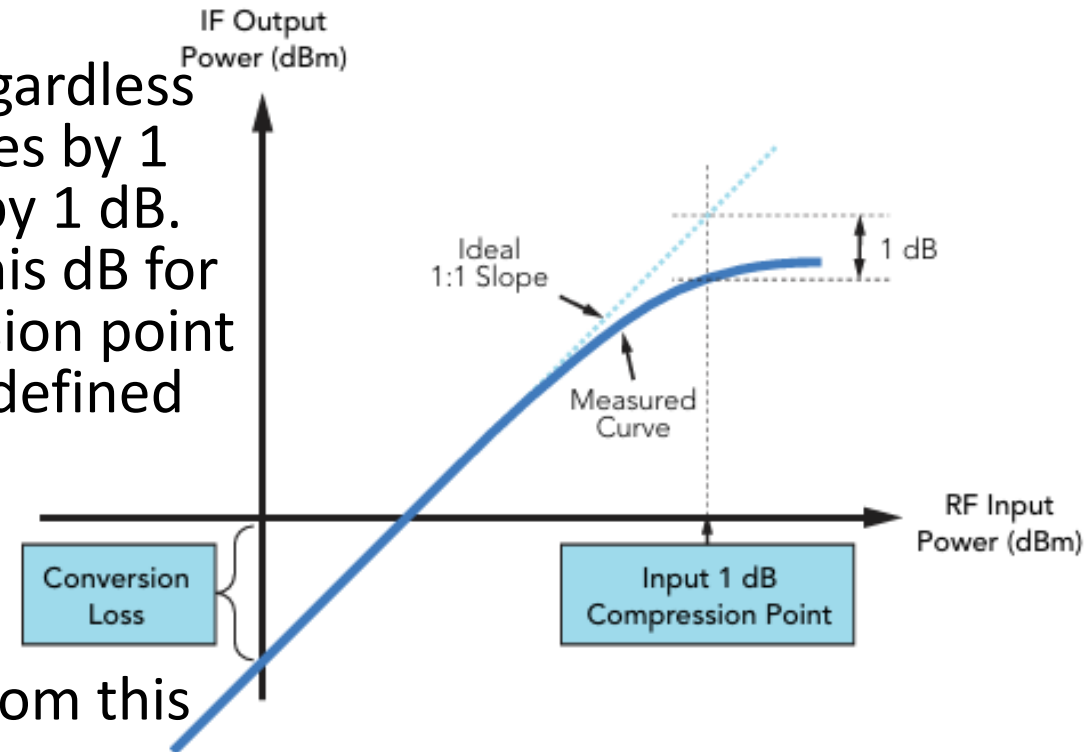
The gain from the RF port to its IF port at specified RF and LO frequencies is called the **conversion gain (or loss)**. Active mixers provide higher conversion gain, and better port-port isolation, but often at the expense of noise and linearity.

Performance metrics

- **Conversion loss:** The most important mixer metric; it is defined as the difference in power between the input RF power level and the desired output IF power level: $CL = P_{RF} - P_{IF}$ with powers expressed in dBm and CL in dB. The theoretically optimum conversion loss for a passive diode mixer is 3.9. Typical values of conversion loss range between about 4.5 to 9 dB. This metric generally shows the quality of the mixer.
- **Isolation:** is a measure of the amount of power that leaks from one mixer port to another. There will always be some small amount of power leakage between the RF, LO and IF ports. Isolation is the difference in power between the input signal and the leaked power to the other ports. The isolation is defined in dB as: $P_{iso} = P_{in(LO)} - P_{out(RF)}$. This quantity is typically reciprocal.
- **1 dB compression:**

Performance metrics

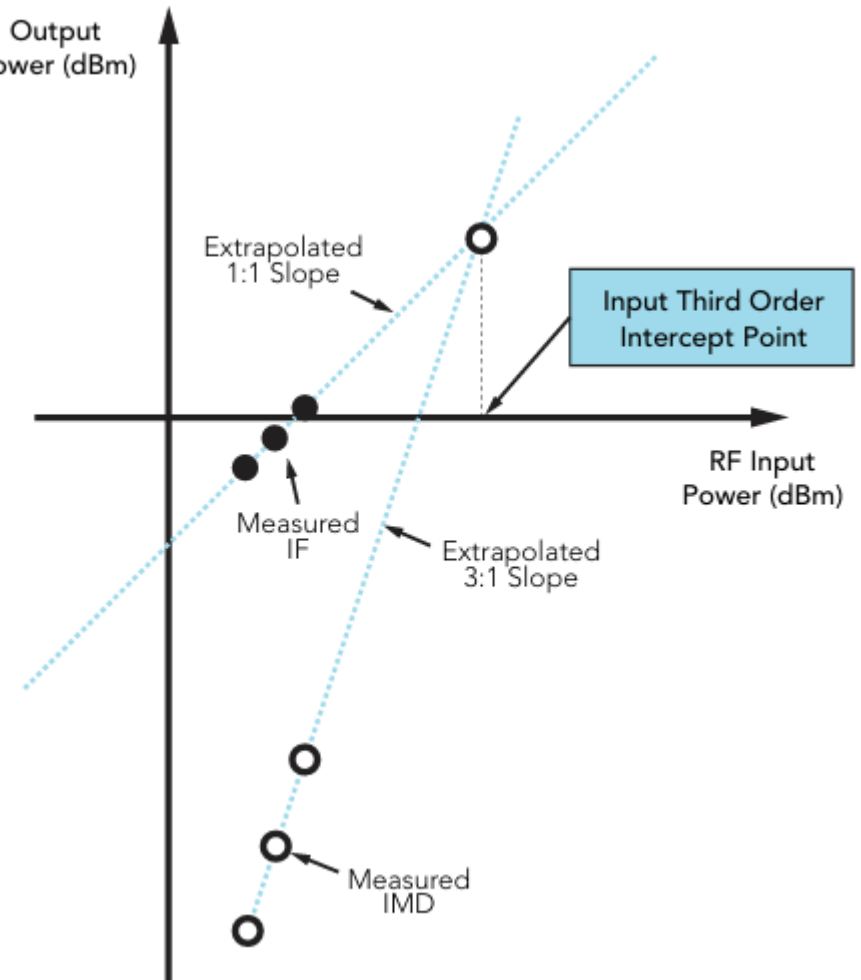
- **1 dB compression:** Under normal operation, the conversion loss of the mixer will be constant, regardless of input RF power. If the input RF power increases by 1 dB, then the output IF power will also increase by 1 dB. However, as the RF power becomes too large, this dB for dB relationship will not hold. The 1 dB compression point is a measure of the linearity of the mixer and is defined as the input RF power required to increase the conversion loss by 1 dB from ideal.



- As the RF power increases, the mixer deviates from this linear behavior and the conversion loss starts to increase. When the input/output curve is reduced by 1 dB (i.e. the conversion loss increases by 1 dB), the input RF 1 dB compression has been reached.
- Conceptually, the 1 dB compression point occurs when the RF signal can no longer be considered “small signal”.

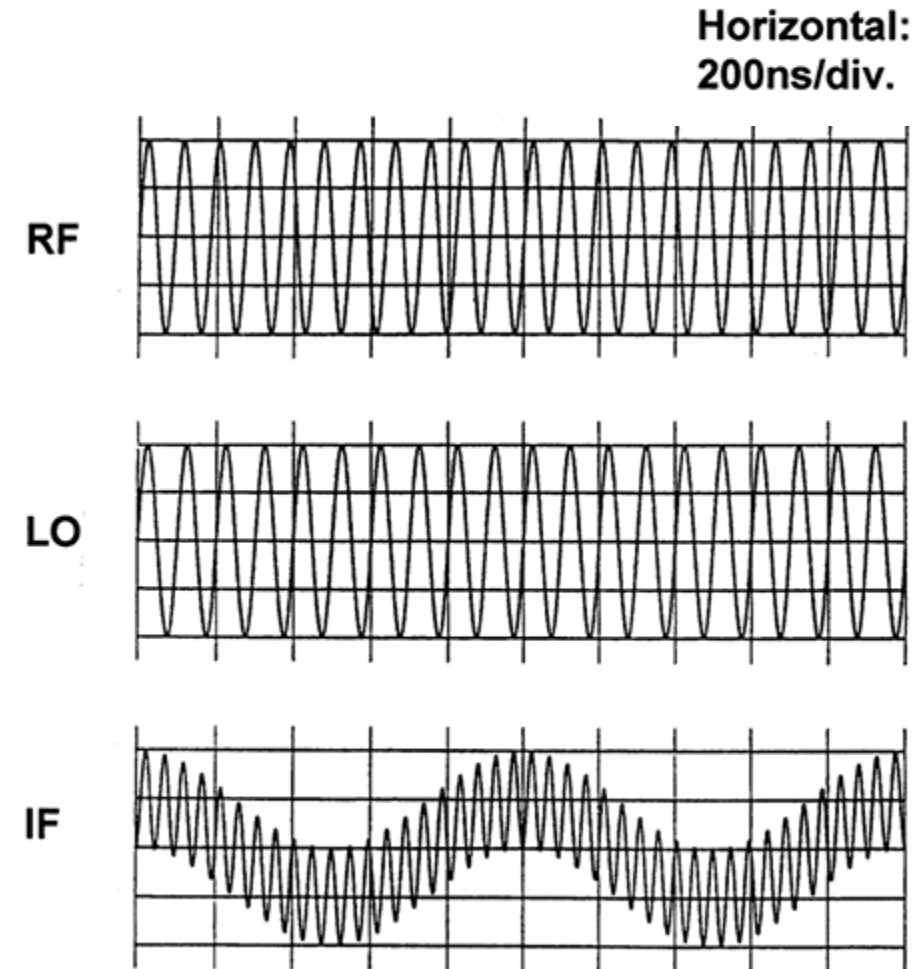
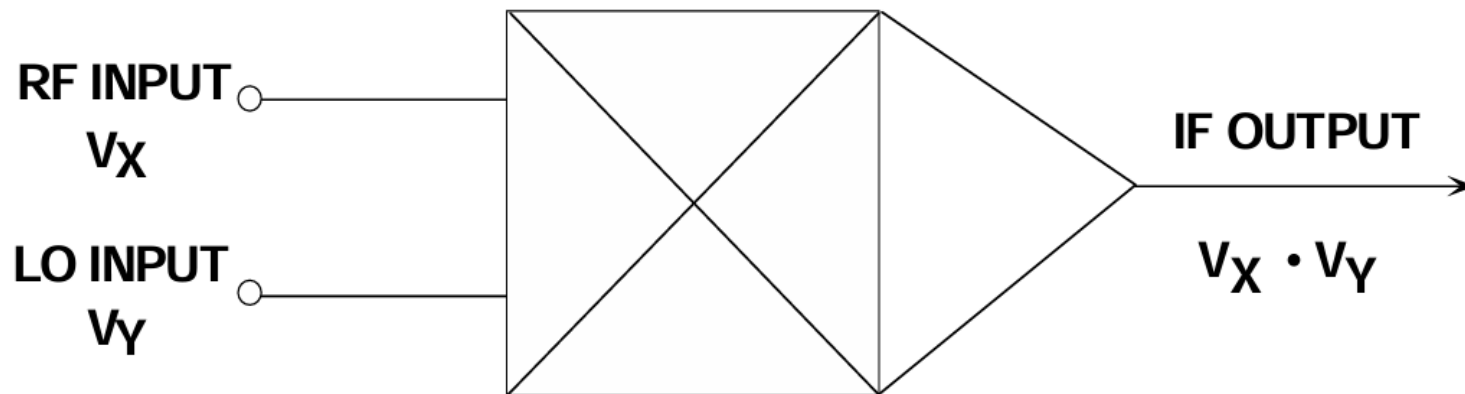
Performance metrics

- **Third order intercept point:** a popular method of determining the “linearity” of a mixer is the "third-order intercept" approach. The Third-Order Intercept Point is a theoretical point on the RF input versus IF output curve where the desired input signal and third-order products become equal in amplitude as RF input is raised.
- As the RF signal increases by 1 dB, the distortion component increases 3 dB. The Third Order Intercept Point is determined by increasing the RF level and noting both the desired and the 3IM levels. The third order intercept point is the point where the extension of the plotted desired output and 3IM output level versus RF input meet. It is not possible to drive the mixer to those RF levels (extrapolation).



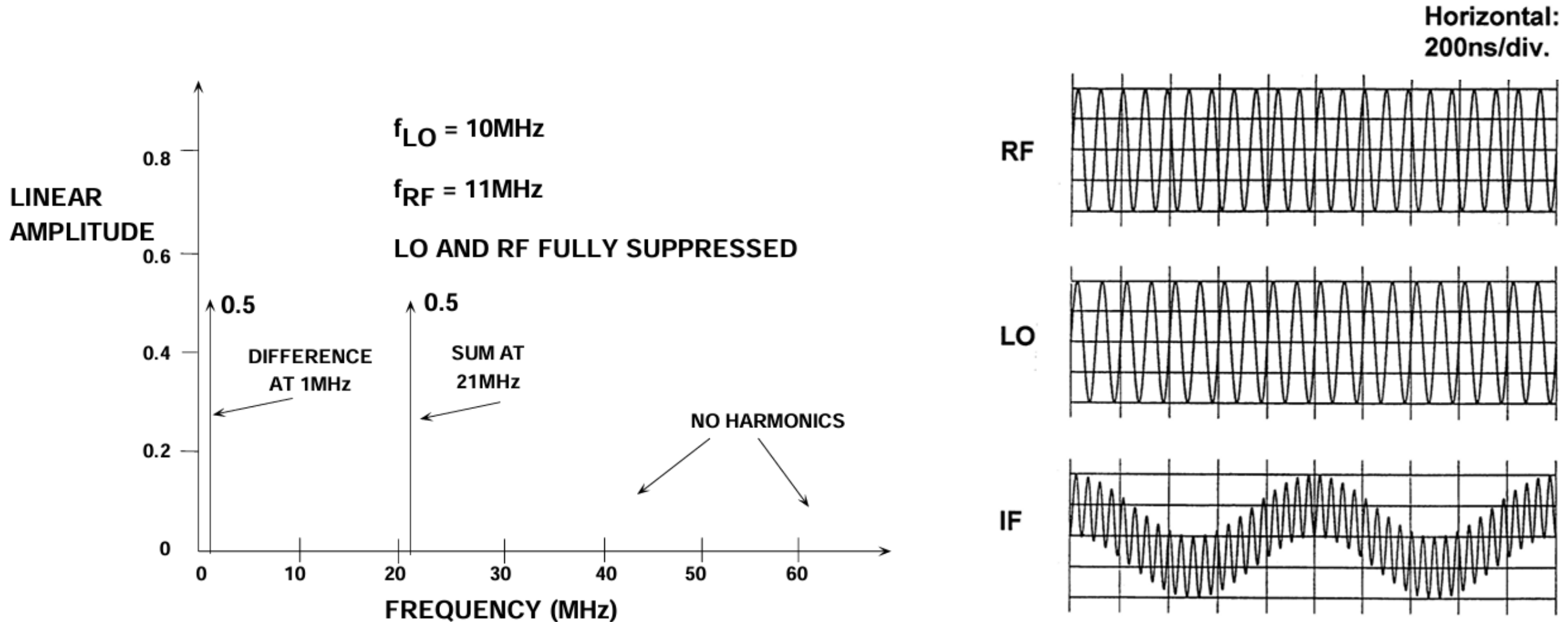
Basic implementation

- Let consider a simplified RF mixer by using an analog multiplier with $f_{RF} = 11MHz$, $f_{LO} = 10MHz$. Figure shows the result of mixing an RF input with a LO input.



Basic implementation

- To better understand mixer behavior, we can consider also the spectrum of the IF output.



Basic implementation

- The relationship is: $\sin \omega_{RF} t \sin \omega_{LO} t = \frac{1}{2} \{ \cos(\omega_{RF} + \omega_{LO}) t + \cos(\omega_{RF} - \omega_{LO}) t \}$
- The multiplier has thus transformed the RF input into two, equal-amplitude cosinusoidal components at its output (the IF port).

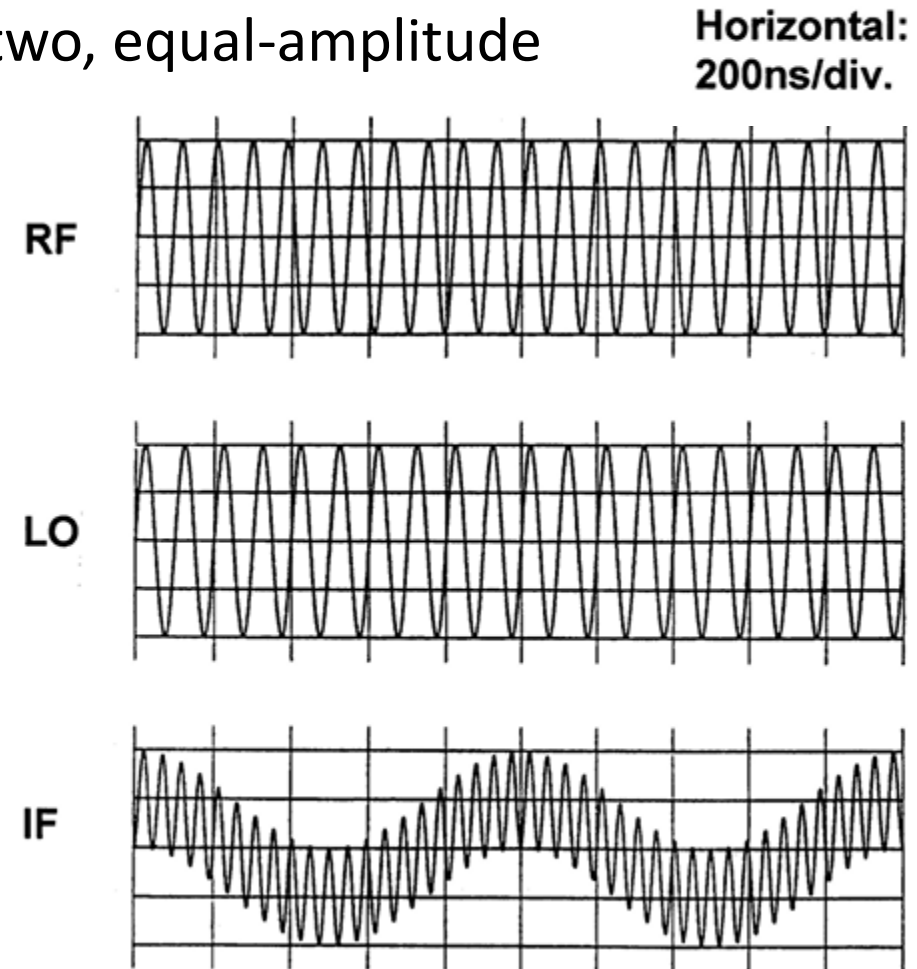
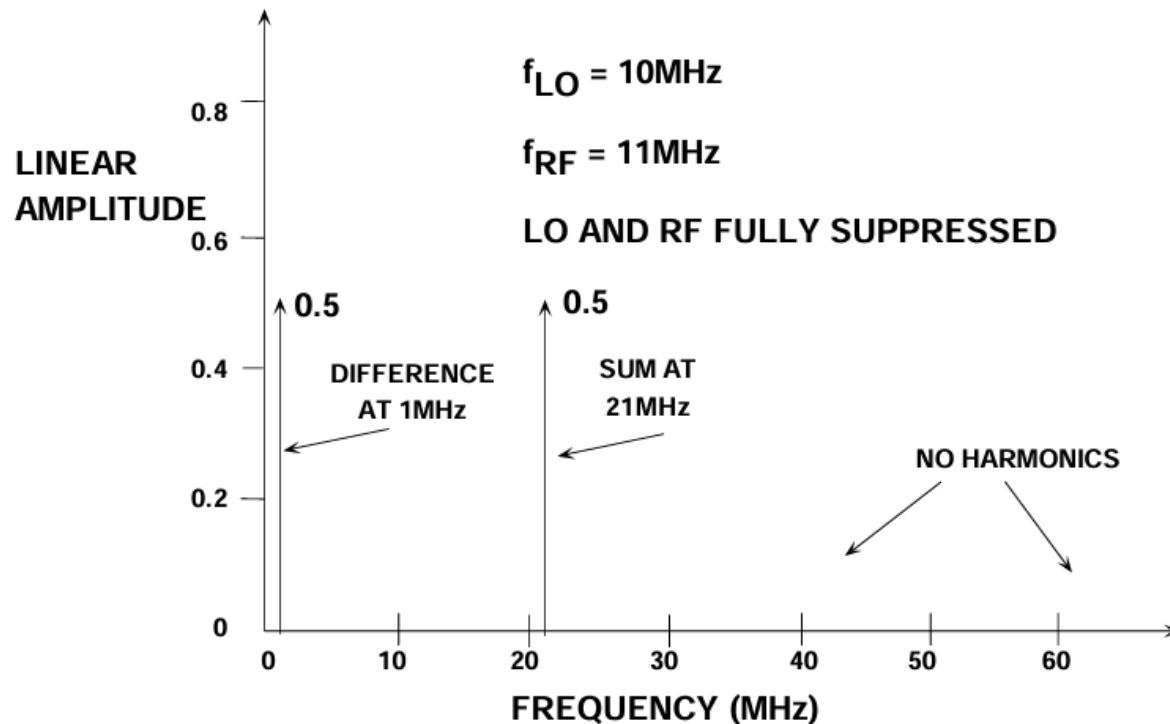
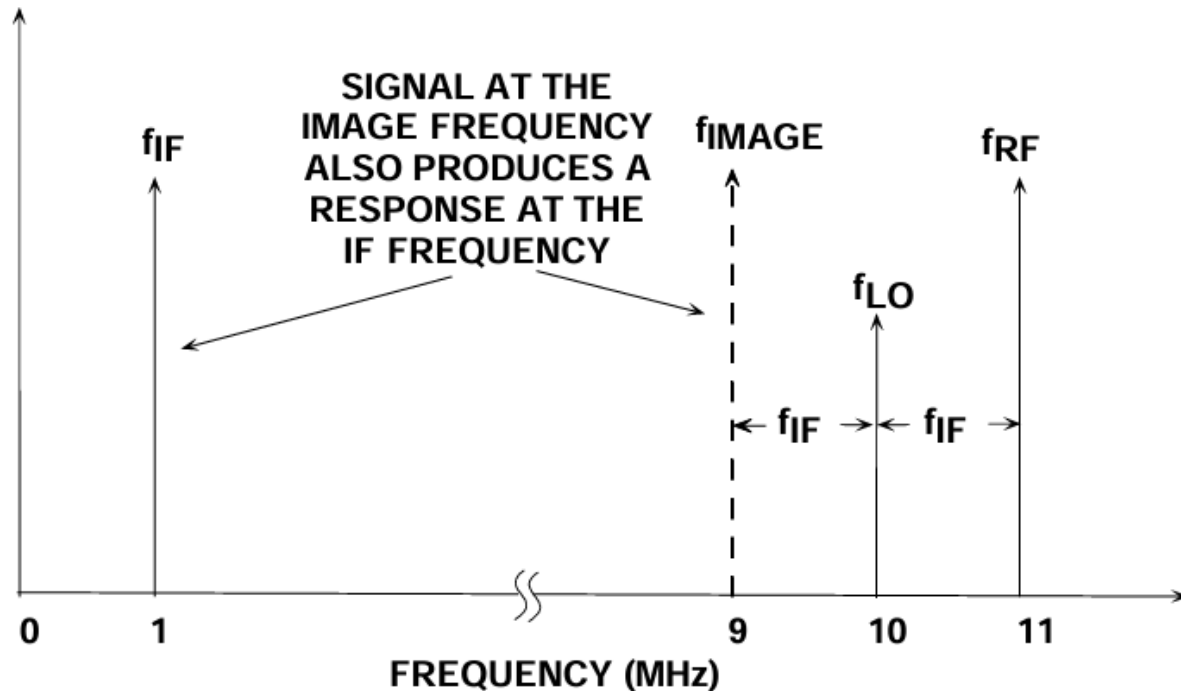


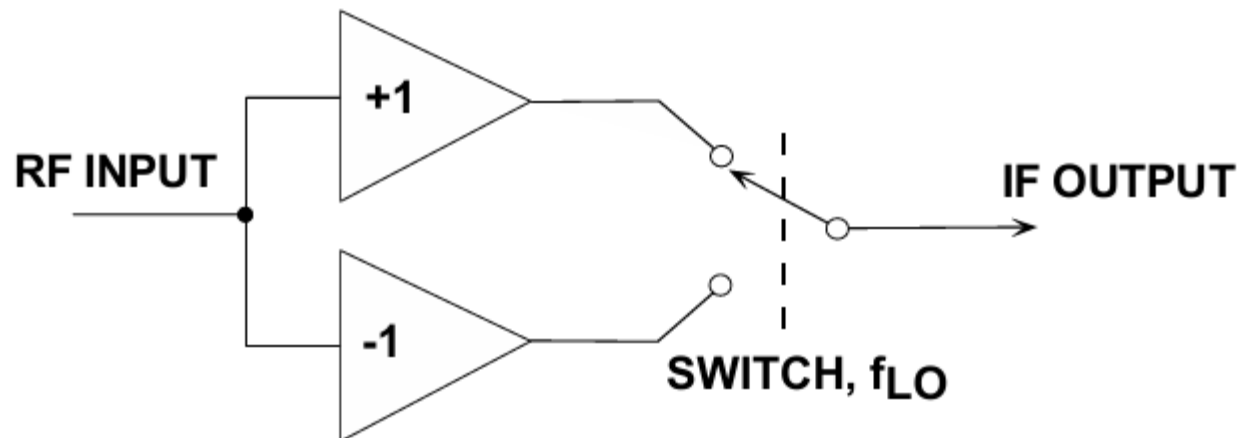
Image response

This mathematically perfect mixer suffers a basic problem, that of image response. The wanted output is found at the difference frequency. However, we have also a spectral component at $\omega_{LO} - \omega_{RF}$ (due to the parity of the cosine function) that is the so-called image frequency that is the same as the IF output but at higher frequency and this image frequency can find its way through the mixer and affect the output.



Basic implementation

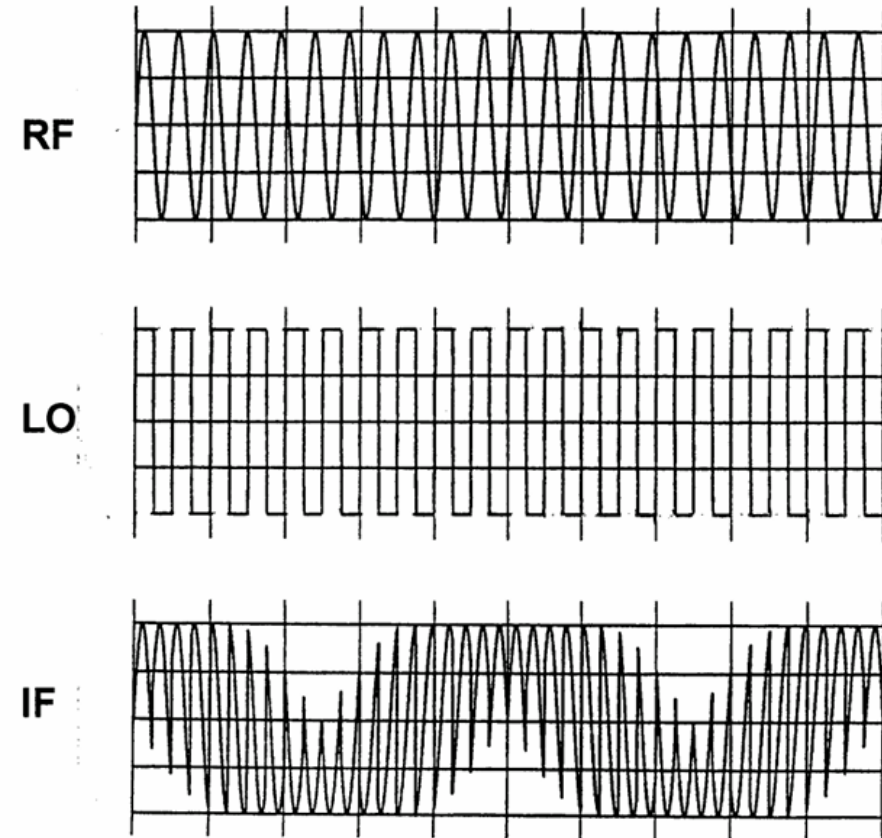
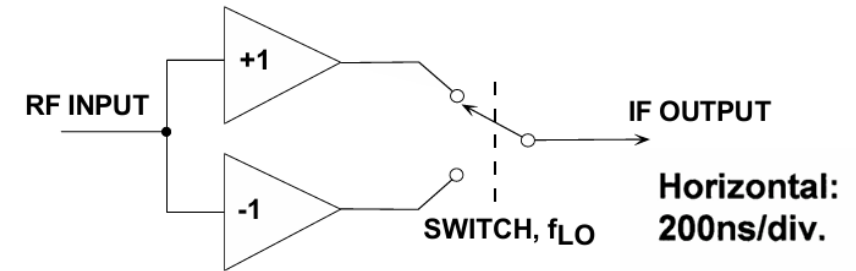
- For an ideal mixer we need some circuit that implements a polarity-switching function in response to the LO input, this type of mixer ensure low-noise and high linearity.
- In this example of ideal mixer, the RF signal is split into in-phase (0°) and anti-phase (180°) components; a changeover switch, driven by the local oscillator (LO) signal, alternately selects the in-phase and antiphase signals (sign-switch).



Basic implementation

- Although simple in concept, the waveform at the intermediate frequency (IF) output can be very complex for even a small number of signals in the input spectrum as shown here (RF input at 11 MHz, and LO of 10 MHz).
- The wanted IF at the difference frequency of 1 MHz is somehow visible in this waveform, also the sum frequency signal at 21 MHz is also apparent.
- In this example we have a product of the input signal: a sinusoid (the RF input) and a square wave (LO input) that can be expressed as a Fourier series as:

$$S_{LO} = \frac{4}{\pi} \left\{ \sin \omega_{LO} t - \frac{1}{3} \sin 3\omega_{LO} t + \frac{1}{5} \sin 5\omega_{LO} t - \dots \right\}$$



Basic implementation

Thus, the output of the mixer is the simple multiplication of the RF and LO as follows:

$$S_{IF} = \frac{4}{\pi} \left\{ \sin \omega_{RF} t \sin \omega_{LO} t - \frac{1}{3} \sin \omega_{RF} t \sin 3\omega_{LO} t + \frac{1}{5} \sin \omega_{RF} t \sin 5\omega_{LO} t - \dots \right\}$$

Expanding each for the products, we obtain:

$$S_{IF} = \frac{2}{\pi} \{ \cos(\omega_{RF} + \omega_{LO})t + \cos(\omega_{RF} - \omega_{LO})t - \frac{1}{3} \cos(\omega_{RF} + 3\omega_{LO})t - \frac{1}{3} \cos(\omega_{RF} - 3\omega_{LO})t + \frac{1}{5} \cos(\omega_{RF} + 5\omega_{LO})t + \frac{1}{5} \cos(\omega_{RF} - 5\omega_{LO})t - \dots \}$$

That can be expressed more simply as:

$$S_{IF} = \frac{2}{\pi} \{ \cos(\omega_{RF} + \omega_{LO})t + \cos(\omega_{RF} - \omega_{LO})t + \textit{harmonics} \}$$

Basic implementation

The most important of these harmonic components are sketched here. Because of the $2/\pi$ term, a mixer has a minimum 3.92 dB insertion loss in the absence of any gain.

The image response is somewhat subtle, as it does not immediately show up in the output spectrum: it is a latent response, awaiting the occurrence of the “wrong” frequency in the input spectrum.

